

Genome-Wide Association Exercise

Data Quality Control

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In this exercise, you will learn how to perform data quality control by removing markers that fail quality control criteria and detecting samples that fail quality control criteria due to amount of missing genotype data. You will also examine your samples for individuals which are related to each other and/or are duplicate samples. Each sample will also be tested for excess homozygosity and heterozygosity of genotype data. Each SNP will be tested for deviations from Hardy-Weinberg Equilibrium. You will also examine QQ plots to see the effect of carrying-out analysis of association study data. These exercises will be carried out using PLINK and R.

1. Running the program:

You can run the program from the dos prompt on your computer or on a LINUX machine or server (highly recommended for larger datasets). In order to get to the folder where the data we are going to analyze is type (please note the data files you will use may be in a different directory and you will be instructed where they are located) :

```
cd plink/exercise/
```

2. Load the data

PLINK can take in your data in many different formats and when you have real data you should examine it and then look over the PLINK documentation (<http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml>) to determine which format is easiest on you to get the data into PLINK. Today's data is already formatted for you in the standard file format. You should have 2 files: a pedfile (GWAS.ped) and a map file (GWAS.map). Please spend some time examining these files and the documentation on the PLINK website before you begin.

Navigate via the command prompt to the directory where your data is found. Then type **plink** in the command prompt and make note of the program output. Next type:

```
plink --file GWAS
```

Note, that PLINK outputs a file called **plink.log** that contains the output of what you see in your command window.

3. Clean the data

Plink outputs information on the number of people and SNPs in your dataset. You now have information to fill in the Oval 1 in the flowchart below.

a. Sample Quality Initial Screen

In the next step of the cleaning process, you will exclude samples that are missing more than 10% of their genotype calls. Such samples are likely to be low quality DNA samples with error-ridden genotype calls.

```
plink --file GWAS --mind 0.10 --recode --out GWAS_clean_mind
```

Look at the file **GWAS_clean_mind.log** to see how many samples are excluded based on this criteria and fill in Box 1.

b. Minor allele frequency, SNP quality and DNA quality criteria:

Type the following:

```
plink --file GWAS_clean_mind --maf 0.05 --recode --out MAF_greater_5
plink --file GWAS_clean_mind --exclude MAF_greater_5.map --recode --out
MAF_less_5
```

Above you are creating two versions of your dataset, one with minor allele frequencies (MAFs) greater than 5% and one with MAFs less than 5%.

You will now clean out SNPs with $MAF > 5\%$ that are missing in more than 5% of the samples and then clean out SNPs with $MAF < 5\%$ that are missing in more than 1% of the samples. Type the following:

```
plink --file MAF_greater_5 --geno 0.05 --recode --out MAF_greater_5_clean
```

Fill in Box 2a.

```
plink --file MAF_less_5 --geno 0.01 --recode --out MAF_less_5_clean
```

Fill in Box 2b.

Finally, in the last line of code you are recombining the cleaned SNPs. Fill Box 2b.

```
plink --file MAF_greater_5_clean --merge MAF_less_5_clean.ped
MAF_less_5_clean.map --recode --out GWAS_MAF_clean
```

Next, we will be more stringent with the sample quality clean by removing samples missing more than 3% of their calls (mind 0.03).

```
plink --file GWAS_MAF_clean --mind 0.03 --recode --out GWAS_clean2
```

You can now fill in Box 3.

c. Check Sex

Entering clinical variables into a database is a tedious process that can be error-prone. It is not always possible to double-check all the variables you have been given. However, you can use the information from the SNP genotypes to verify the sex of individuals in your study. This is done by looking at the homozygosity (F) on the X chromosome in each individual. This number is expected to be less than 0.2 in females and greater than 0.8 in males.

Question 1:

Why do you expect the homozygosity rate to be higher on the X chromosome in males than females?

Because males only have one X chromosome. The lack of extra alleles from a second X chromosome means they will have a higher homozygous rate.

Run the following command in plink:

```
plink --file GWAS_clean2 --check-sex --out GWAS_sex_checking
```

Use R to open the file GWAS_sex_checking.sexcheck and determine if there are any individuals who may have sex coded incorrectly.

```
sexcheck = read.table("GWAS_sex_checking.sexcheck", header=T)
names(sexcheck)
sex_problem = sexcheck[which(sexcheck$STATUS=="PROBLEM"),]
sex_problem
```

From that file, NA20530 and NA20506 was coded as a female (sex code = 2) and from the genotypes appears to be males (sex code = 1). In addition, 3 individuals (NA20766, NA20771 and NA20757) do not have enough information for PLINK to decide if they are males or females so the program reports sex = 0 for the genotyped sex. Fill in the table below:

Table 1: Sex check

FID	IID	PEDSEX	SNPSEX	STATUS	F
NA20506	NA20506	2	1	PROBLEM	1.000
NA20530	NA20530	2	1	PROBLEM	1.000
NA20766	NA20766	2	0	PROBLEM	0.2141
NA20771	NA20771	2	0	PROBLEM	0.2292
NA20757	NA20757	2	0	PROBLEM	0.2234

There are two reasons for these kinds of discrepancies. The first is that the records are wrong. So you would ask your collaborators to double-check the sex in the clinical records. The second reason is that you may not have genotyped enough SNPs on the X chromosome to predict the sex. In this dataset, there are 194 SNPs on the X chromosome.

Question 2:

With a limited number of SNPs genotyped on the X chromosome, are you concerned PLINK is indicating that some females are males? What does this mean for the interpretation of your results?

The small number of SNPs could lead to inaccuracies. To avoid misinterpretation, flagged results should be cross-checked with collaborators and other data.

d. Duplicates

Your collaborator has informed you that individual NA25001 in family 25985 is a duplicate of individual NA12057 in family 1344. Use the following command in PLINK to check if there are any duplicate samples in the dataset:

```
plink --file GWAS_clean2 --genome --out duplicates
```

In larger datasets (more people and more markers) this command takes a lot longer. It is calculating an IBS matrix between all members of the study. However, the genome command needs to be run only once and in subsequent analyses the read-genome command can be used to access this information.

Open the **duplicates.genome** file R with the following command:

```
dups = read.table("duplicates.genome", header = T)
```

We are interested in the value for Pi-Hat (the proportion IBD or in this dataset IBS). You may notice that you have more than one duplicate. Also, keep your eyes open for individuals who have Pi-Hat values that are high.

```
problem_pairs = dups[which(dups$PI_HAT > 0.4),]  
problem_pairs
```

Table 2: Duplicates and relatedness

FID1	IID1	FID2	IID2	PI_HAT
1344	NA12057	13291	NA25001	0.9988
1444	NA12739	1444	NA12748	0.5000
1444	NA12739	1444	NA12749	0.5071
M033	NA19774	M041	NA25000	1.0000

Question 3: How many duplicate pairs do you find (**hint: Pi-Hat = 1**)? What proportion would you expect a parent/ child to share IBS? Can you find any such relationship?

We can see one duplicate pair (Pi-Hat=1).

We would expect a child to be around a half match to their parent (Pi-Hat=0.5).

There is also a child-parent pair.

Word of caution: Pi-hat can be inflated and many individuals appear to be related to each other if you have samples from different population. This explains why we see a number of pairs of individuals with Pi-hat greater than 0.05 since three distinct populations were sampled.

Additionally this phenomenon can be observed if a subset of your sample was genotyped using "bad" chips, which creates two or more populations and the individuals within these populations appear to be more closely related than the true situation.

In R see for yourself how many pairs have pi-hat great than 0.05 with the following code:

```
problem_pairs = dups[which(dups$PI_HAT > 0.05),]  
myvars = c("FID1", "IID1", "FID2", "IID2", "PI_HAT")  
problem_pairs[myvars]
```

Make a txt file that looks like this:

```
1344 NA12057  
1444 NA12739  
M033 NA19774
```

name it 'IBS_excluded.txt' saving it in the folder with your PLINK data. Then type the command :

```
plink --file GWAS_clean2 --remove IBS_excluded.txt --recode --out GWAS_clean3
```

You can now fill in Box 4 and Oval 3.

Question 4:

Your collaborators have given you a list of duplicates in the dataset that were sent as quality control for genotyping. You have this list so why would you bother obtaining the IBS matrix on all the samples in your study?

It is another way to confirm our data. Our collaborators might have made a missed a duplicate or made a mistake. Furthermore, it helps us identify relatedness among our samples.

e. Excess homozygosity and heterozygosity

Type the following:

```
plink --file GWAS_clean3 --het
```

and then open up R and examine the data for excess homozygosity and heterozygosity. Open up the R program by typing:

R

then in R type:

```
Dataset <- read.table("plink.het", header=TRUE, sep="", na.strings="NA",  
dec=".", strip.white=TRUE)
```

```
mean(Dataset$F)  
sd(Dataset$F)
```

```
jpeg("hist.jpeg", height=1000, width=1000)  
hist(scale(Dataset$F), xlim=c(-4,4))  
dev.off()
```

in order to obtain the mean and SD of the inbreeding coefficient (F) for each person in the study and additionally to plot a histogram so you can observe if there are any outliers. The last command should create a new jpeg image file (filename=hist.jpeg) in your working directory. You should be concerned if you have individuals who are greater than 4 SDs from the mean of F. F is calculated based on the observed number of homozygous genotype calls an individual has in comparison with the expected number of homozygous genotype calls. So if an individual has less homozygous call than expected F is negative and if the individual has more homozygous calls than expected F is positive.

Those individuals whose genotypes either have excess homozygosity (more homozygous calls than expected... F is positive) or excess heterozygosity (less homozygous calls than expected... F is negative) if F greater than 4SD from the mean F and these samples should be removed from the dataset. In our case, you can see from the histogram that we do not have this problem. Fill this information in Box 5.

Question 5:

You observe a sample with a negative value for F that is 6 SD outside the mean F for the dataset. Is this excess heterozygosity or excess homozygosity? Give one reason a sample might exhibit this.

Negative samples correlate with heterozygosity, while positive ones homozygosity

We can conclude the sample in question has excess heterozygosity.

This could be an exaple of ancestry from genetically divergant populations.

f. Hardy-Weinberg Equilibrium (HWE):

This is the first time we are introducing a trait in the cleaning. You should note that all of the cleaning above took place with the phenotype as missing in the ped file. Type the following to obtain an output of HWE test for all SNPs for the trait Aff:

```
plink --file GWAS_clean3 --pheno pheno.txt --pheno-name Aff --hardy
```

Open the file **plink.hwe** and look for SNPs with p-values of 10^{-7} or smaller.

In R:

```
hardy = read.table("plink.hwe", header = T)
names(hardy)
hwe_prob = hardy[which(hardy$P < 0.0000009),]
hwe_prob
```

Using a criteria of p of 10^{-7} or smaller to for HWE to be out, how many SNPs fail HWE in the controls? Fill out Oval 5 and Box 4. Using the same criteria, how many SNPs fail HWE in the cases? However, we will not exclude based on this criteria in the cases. We will only make a note of these SNPs. Are there any SNPs that fail HWE using this criterion in the cases? Complete Table 2 with this information.

Table 3: Hardy-Weinberg Equilibrium

Cases		Controls	
SNP	Pvalue	SNP	Pvalue
rs2968487	3.877e-10	rs2968487	2.262e-07
rs11239985	7.625e-08		
rs12988076	6.973e-07		

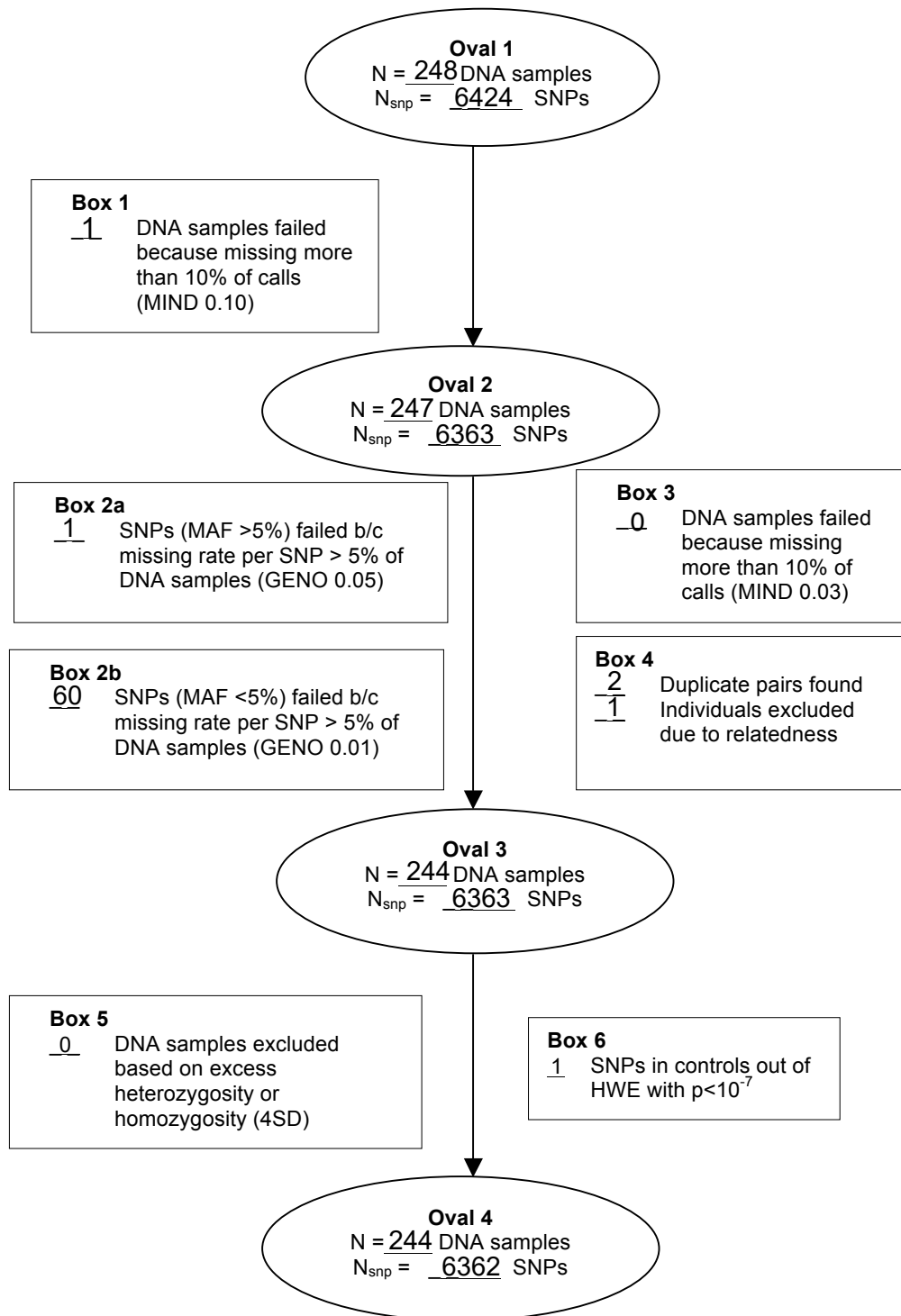
Create a text file called HWE_out.txt with the following SNP in it:

rs2968487

and then type the following for PLINK:

```
plink --file GWAS_clean3 --exclude HWE_out.txt --recode --out GWAS_clean4
```

There are a number of SNPs with HWE pvalues in the range of 10^{-5} to 10^{-6} in the controls and in the cases. Based on above criteria they will not be excluded however, a note should be made for such SNPs and kept in mind if they reach genome-wide significance during association testing. You can now fill in Box 6 and Oval 4.



Solutions to Questions:

Oval 1 and 2 also and Box 1 information:

Analysis finished: Thu Jul 7 13:25:11 2011

```
@-----@
|          PLINK!          |          v1.07          |          10/Aug/2009          |
|-----|
| (C) 2009 Shaun Purcell, GNU General Public License, v2 |
|-----|
| For documentation, citation & bug-report instructions: |
|          http://pngu.mgh.harvard.edu/purcell/plink/      |
|-----@
```

Web-based version check (--noweb to skip)
Recent cached web-check found...Problem connecting to web

Writing this text to log file [GWAS_clean_mind.log]
Analysis started: Thu Jul 7 13:26:33 2011

Options in effect:
--file GWAS
--mind 0.10
--recode
--out GWAS_clean_mind

6424 (of 6424) markers to be included from [GWAS.map] [Oval 1]
248 individuals read from [GWAS.ped] [Oval 1]

0 individuals with nonmissing phenotypes
Assuming a disease phenotype (1=unaff, 2=aff, 0=miss)
Missing phenotype value is also -9
0 cases, 0 controls and 248 missing
125 males, 123 females, and 0 of unspecified sex
Before frequency and genotyping pruning, there are 6424 SNPs
248 founders and 0 non-founders found
Writing list of removed individuals to [GWAS_clean_mind.irem]
1 of 248 individuals removed for low genotyping (MIND > 0.1) [Box 1]
6 heterozygous haploid genotypes; set to missing
Writing list of heterozygous haploid genotypes to [GWAS_clean_mind.hh]
1 SNPs with no founder genotypes observed
Warning, MAF set to 0 for these SNPs (see --nonfounders)
Writing list of these SNPs to [GWAS_clean_mind.nof]
Total genotyping rate in remaining individuals is 0.996863
0 SNPs failed missingness test (GENO > 1)
0 SNPs failed frequency test (MAF < 0)

After frequency and genotyping pruning, there are 6424 SNPs [Oval 2]
After filtering, 0 cases, 0 controls and 247 missing [Oval 2]
After filtering, 125 males, 122 females, and 0 of unspecified sex
Writing recoded ped file to [GWAS_clean_mind.ped]
Writing new map file to [GWAS_clean_mind.map]

Analysis finished: Thu Jul 7 13:26:34 2011

Box 2a information:

```
@-----@
|          PLINK!          |          v1.07          |          10/Aug/2009          |
|-----|
| (C) 2009 Shaun Purcell, GNU General Public License, v2 |
|-----|
| For documentation, citation & bug-report instructions: |
|          http://pngu.mgh.harvard.edu/purcell/plink/      |
|-----@
```

Web-based version check (--noweb to skip)
Recent cached web-check found...Problem connecting to web

Writing this text to log file [MAF_greater_5_clean.log]
Analysis started: Thu Jul 7 13:28:30 2011

Options in effect:
--file MAF_greater_5
--geno 0.05
--recode
--out MAF_greater_5_clean

5867 (of 5867) markers to be included from [MAF_greater_5.map]
247 individuals read from [MAF_greater_5.ped]
0 individuals with nonmissing phenotypes
Assuming a disease phenotype (1=unaff, 2=aff, 0=miss)
Missing phenotype value is also -9
0 cases, 0 controls and 247 missing
125 males, 122 females, and 0 of unspecified sex
Before frequency and genotyping pruning, there are 5867 SNPs
247 founders and 0 non-founders found
6 heterozygous haploid genotypes; set to missing
Writing list of heterozygous haploid genotypes to [MAF_greater_5_clean.hh]
Total genotyping rate in remaining individuals is 0.997028
1 SNPs failed missingness test (GENO > 0.05) [Box 2a]
0 SNPs failed frequency test (MAF < 0)
After frequency and genotyping pruning, there are 5866 SNPs
After filtering, 0 cases, 0 controls and 247 missing
After filtering, 125 males, 122 females, and 0 of unspecified sex
Writing recoded ped file to [MAF_greater_5_clean.ped]
Writing new map file to [MAF_greater_5_clean.map]

Analysis finished: Thu Jul 7 13:28:31 2011

Box 2b information:

```
@-----@
|          PLINK!          |          v1.07          |          10/Aug/2009          |
|-----|
| (C) 2009 Shaun Purcell, GNU General Public License, v2 |
|-----|
| For documentation, citation & bug-report instructions: |
|          http://pngu.mgh.harvard.edu/purcell/plink/      |
|-----|
@-----@
```

Web-based version check (--noweb to skip)
Recent cached web-check found...Problem connecting to web

Writing this text to log file [MAF_less_5_clean.log]
Analysis started: Thu Jul 7 13:32:15 2011

Options in effect:
--file MAF_less_5
--geno 0.01
--recode
--out MAF_less_5_clean

557 (of 557) markers to be included from [MAF_less_5.map]
247 individuals read from [MAF_less_5.ped]
0 individuals with nonmissing phenotypes
Assuming a disease phenotype (1=unaff, 2=aff, 0=miss)
Missing phenotype value is also -9
0 cases, 0 controls and 247 missing
125 males, 122 females, and 0 of unspecified sex
Before frequency and genotyping pruning, there are 557 SNPs
247 founders and 0 non-founders found
Total genotyping rate in remaining individuals is 0.995123

60 SNPs failed missingness test (GENO > 0.01) [Box 2b]

0 SNPs failed frequency test (MAF < 0)

After frequency and genotyping pruning, there are 497 SNPs

After filtering, 0 cases, 0 controls and 247 missing

After filtering, 125 males, 122 females, and 0 of unspecified sex

Writing recoded ped file to [MAF_less_5_clean.ped]

Writing new map file to [MAF_less_5_clean.map]

Analysis finished: Thu Jul 7 13:32:15 2011

Oval 2 and Box 3 information:

```
@-----@
|          PLINK!          |          v1.07          |          10/Aug/2009          |
|-----|
| (C) 2009 Shaun Purcell, GNU General Public License, v2 |
|-----|
| For documentation, citation & bug-report instructions: |
|          http://pngu.mgh.harvard.edu/purcell/plink/      |
|-----@
```

Web-based version check (--noweb to skip)

Recent cached web-check found...Problem connecting to web

Writing this text to log file [GWAS_clean2.log]

Analysis started: Thu Jul 7 15:03:38 2011

Options in effect:

--file GWAS_MAF_clean

--mind 0.03

--recode

--out GWAS_clean2

6363 (of 6363) markers to be included from [GWAS_MAF_clean.map]

247 individuals read from [GWAS_MAF_clean.ped]

0 individuals with nonmissing phenotypes

Assuming a disease phenotype (1=unaff, 2=aff, 0=miss)

Missing phenotype value is also -9

0 cases, 0 controls and 247 missing

125 males, 122 females, and 0 of unspecified sex

Before frequency and genotyping pruning, there are 6363 SNPs

247 founders and 0 non-founders found

0 of 247 individuals removed for low genotyping (MIND > 0.03) [Box 3]

6 heterozygous haploid genotypes; set to missing

Writing list of heterozygous haploid genotypes to [GWAS_clean2.hh]

Total genotyping rate in remaining individuals is 0.99716

0 SNPs failed missingness test (GENO > 1)

0 SNPs failed frequency test (MAF < 0)

After frequency and genotyping pruning, there are 6363 SNPs

After filtering, 0 cases, 0 controls and 247 missing

After filtering, 125 males, 122 females, and 0 of unspecified sex

Writing recoded ped file to [GWAS_clean2.ped]

Writing new map file to [GWAS_clean2.map]

Analysis finished: Thu Jul 7 15:03:40 2011

Answer to Question 1:

Why do you expect the homozygosity rate to be higher on the X chromosome in males than females?

Because males only have one allele for each SNP on the X chromosome they will appear homozygous.

Table 1: Sex check

FID	IID	PEDSEX	SNPSEX	STATUS	F
NA20506	NA20506	2	1	PROBLEM	1
NA20530	NA20530	2	1	PROBLEM	1
NA20766	NA20766	2	0	PROBLEM	0.2292
NA20771	NA20771	2	0	PROBLEM	0.2234
NA20757	NA20757	2	0	PROBLEM	0.2141

Answer to Question 2:

With a limited number of SNPs genotyped on the X chromosome, are you concerned PLINK is indicating that some females are males? What does this mean for the interpretation of your results?

No. As a result of the lack of information, PLINK may indicate that the genotyped sex is male when in fact it is female (think back to the question 4 answer). In our case, our collaborators confirmed that the records are correct. We will not be changing the sex of any individual based on our analysis because we do not have enough information to support changing the sex code.

Table 2: Duplicates and relatedness

FID1	IID1	FID2	IID2	PI_HAT
M041	NA25000	M033	NA19774	1
13291	NA25001	1344	NA12057	1.00
1444	NA12739	1444	NA12749	0.51
1444	NA12739	1444	NA12748	0.50

Answer to Question 3:

How many duplicate pairs do you find (**hint: Pi-Hat = 1**)? What proportion would you expect a parent/ child to share IBS? Can you find any such relationship? Fill in Box 4.

You have one surprise duplicate pair (NA25001 and NA12057). You would expect a parent/child relationship to have a Pi Hat value of 0.5. So NA12749 and NA12748 are the parents of NA12739 and you should exclude NA12739 in addition to NA19774 and NA12057 because they are duplicate samples of NA25000 and NA25001, respectively.

Oval 3 information

```
@-----@
|          PLINK!          |          v1.07          |          10/Aug/2009          |
|-----|
| (C) 2009 Shaun Purcell, GNU General Public License, v2 |
|-----|
| For documentation, citation & bug-report instructions: |
|          http://pngu.mgh.harvard.edu/purcell/plink/      |
|-----@
```

Web-based version check (--noweb to skip)
Recent cached web-check found...Problem connecting to web

Writing this text to log file [GWAS_clean3.log]
Analysis started: Thu Jul 7 16:21:56 2011

Options in effect:
--file GWAS_clean2
--remove IBS_excluded.txt
--recode
--out GWAS_clean3

6363 (of 6363) markers to be included from [GWAS_clean2.map]
247 individuals read from [GWAS_clean2.ped]
0 individuals with nonmissing phenotypes
Assuming a disease phenotype (1=unaff, 2=aff, 0=miss)
Missing phenotype value is also -9
0 cases, 0 controls and 247 missing
125 males, 122 females, and 0 of unspecified sex
Reading individuals to remove [IBS_excluded.txt] ... 3 read
3 individuals removed with --remove option
Before frequency and genotyping pruning, there are 6363 SNPs
244 founders and 0 non-founders found
6 heterozygous haploid genotypes; set to missing
Writing list of heterozygous haploid genotypes to [GWAS_clean3.hh]
Total genotyping rate in remaining individuals is 0.997225
0 SNPs failed missingness test (GENO > 1)
0 SNPs failed frequency test (MAF < 0)
After frequency and genotyping pruning, there are 6363 SNPs [Oval 3]
After filtering, 0 cases, 0 controls and 244 missing [Oval 3]
After filtering, 123 males, 121 females, and 0 of unspecified sex
Writing recoded ped file to [GWAS_clean3.ped]
Writing new map file to [GWAS_clean3.map]

Analysis finished: Thu Jul 7 16:21:58 2011

Answer to Question 4:

Your collaborators have given you a list of duplicates in the dataset that were sent as quality control for genotyping. You have this list so why would you bother obtaining the IBS matrix on all the samples in your study?

You need to thoroughly check the integrity of your dataset. Yes, you want to know if the expected duplicates are correct but you are also wondering if another sample was accidentally sent as a duplicate. The common example of this is when clinical investigators in a large study recruit the same patient at two different time points. The same person is given two different sample IDs.

Answer to Question 5:

You observe a sample with a negative value for F that is 6 SD outside the mean F for the dataset. Is this excess heterozygosity or excess homozygosity? Give one reason a sample might exhibit this.

Excess heterozygosity. The sample has less homozygous calls than expected. One reason is that the sample could be contaminated with DNA from another person. Maybe DNA was transferred to the same tube twice from two different people?

Table 3: Hardy Weinberg Equilibrium

Fail Cases		Fail Controls	
SNP	pvalue	SNP	pvalue
None		rs2968487	2.262e-007

```
@-----@
|          PLINK!          |          v1.07          |    10/Aug/2009    |
|-----|
| (C) 2009 Shaun Purcell, GNU General Public License, v2 |
|-----|
| For documentation, citation & bug-report instructions: |
|      http://pngu.mgh.harvard.edu/purcell/plink/         |
|-----|
@-----@
```

Web-based version check (--noweb to skip)
Recent cached web-check found...Problem connecting to web

Writing this text to log file [GWAS_clean4.log]
Analysis started: Thu Jul 7 16:26:18 2011

Options in effect:
--file GWAS_clean3
--exclude HWE_out.txt
--recode
--out GWAS_clean4

6363 (of 6363) markers to be included from [GWAS_clean3.map]
244 individuals read from [GWAS_clean3.ped]
0 individuals with nonmissing phenotypes
Assuming a disease phenotype (1=unaff, 2=aff, 0=miss)
Missing phenotype value is also -9
0 cases, 0 controls and 244 missing
123 males, 121 females, and 0 of unspecified sex
Reading list of SNPs to exclude [HWE_out.txt] ... 1 read
Before frequency and genotyping pruning, there are 6362 SNPs
244 founders and 0 non-founders found
6 heterozygous haploid genotypes; set to missing
Writing list of heterozygous haploid genotypes to [GWAS_clean4.hh]
Total genotyping rate in remaining individuals is 0.997229
0 SNPs failed missingness test (GENO > 1)
0 SNPs failed frequency test (MAF < 0)
After frequency and genotyping pruning, there are 6362 SNPs [Oval 4]
After filtering, 0 cases, 0 controls and 244 missing [Oval 4]
After filtering, 123 males, 121 females, and 0 of unspecified sex
Writing recoded ped file to [GWAS_clean4.ped]
Writing new map file to [GWAS_clean4.map]

Analysis finished: Thu Jul 7 16:26:20 2011

